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SIMATS Online Programming Lab
PYTHON PROGRAMMING

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Q1 Sum the given number upto single digit 5 marks

Write a Python code to find the sum of the given number upto single digit
input: 89789
output: 5

Your Code

```
1 n=int(input())
2 while n>=10:
3     s=0
4     while n>0:
5         s+=n%10
6         n//=10
7     n=s
8 print(n)
```

Input
89789

Output
5

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Q2 Prime Number 5 marks

Find the next prime number
input: 13
output: 17

Your Code

```
1 n=int(input())
2 while True:
3     n+=1
4     prime=True
5     for i in range(2,n):
6         if n%i==0:
7             prime=False
8             break
9     if prime:
10        print(n)
11        break
```

Input
13

Output
17

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Q3 Swap the first and last digits 5 marks

WAP to swap the first and last digits of the given number.
Input: 23546
Output: 63542

Your Code

```
1 n=input()
2 if len(n)>1:
3     n=n[-1]+n[1:-1]+n[0]
4 print(n)
```

Input
23546

Output
63542

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Q4 Odd Sequence first and Even Sequence next 5 marks

WAP to add Sequence first and Even Sequence next
Input: 658134
Output: 513684

Your Code

```
1 n=input()
2 odd=""
3 even=""
4 for i in n:
5     if int(i)%2==0:
6         even+=i
7     else:
8         odd+=i
9 print(odd+even)
```

Input
658134

Output
513684

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WAP to check the given number is step number or not
A step number is a number in which the difference between every pair of consecutive digits is 1.
For example:
54567 → Yes
[5-4] = 1
[4-5] = 1
[5-6] = 1
[6-7] = 1
1235 → No
[3-5] = 2

Your Code

```
1 n=input()
2 step=True
3 for i in range(len(n)-1):
4     if abs(int(n[i])-int(n[i+1]))!=1:
5         step=False
6         break
7 if step:
8     print("Yes")
```

Input
54567

Output
Yes

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WAP to reverse only the even numbers
Input: 325649
Output: 345629

Your Code

```
1 n=input()
2 even=[]
3 for i in n:
4     if int(i)%2==0:
5         even.append(i)
6 even.reverse()
7 result=""
8 for i in n:
9     if int(i)%2==0:
10        result+=even.pop()
11    else:
12        result+=i
13 print(result)
```

Input
325649

Output
345629

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WAP to encode the given number
Input: 55553322255
Output: 54322352

Your Code

```
1 n=input()
2 ans=""
3 count=1
4 for i in range(1,len(n)):
5     if n[i]==n[i-1]:
6         count+=1
7     else:
8         ans += n[i-1]+str(count)
9         count = 1
10 ans += n[-1] + str(count)
11 print(ans)
```

Input
55553322255

Output
54322352

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Q5 Reverse Odd Digit 5 marks

WAP to Reverse Odd Digits Only
Input: 816253
Output: 351

Your Code

```
1 n=input()
2 result=""
3 for i in n:
4     if int(i)%2!=0:
5         result = i+result
6 print(result)
```

Input
816253

Output
351

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Q9 Convert every number into the next sequence number.

5 marks

WAP to convert every number into the next sequence number.

Input: 89781

Output: 90892

Your Code

```
1 n=input()
2 result=""
3 for i in n:
4     if i=='9':
5         result=result+'0'
6     else:
7         result=result+str(int(i)+1)
8 print(result)
```

Input

89781

Output

90892

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Q10 Print the total factor count of a given number

5 marks

WAP to print the total factor count of a given number

Input: 12

Output: 6

Your Code

```
1 n=int(input())
2 count=0
3 for i in range(1,n+1):
4     if n%i==0:
5         count +=1
6 print(count)
```

Input

12

Output

6